



Course Outline

1. Course Identity

The information in this block should be exactly as approved by CUHK Senate. In case there are any differences, please explain in the table below.

Course code	MAT2042
Course title (English)	Honours Linear Algebra
Course title (Chinese)	线性代数荣誉课程
Units	3
Language of Instruction	English
Description (English)	This course introduces the fundamental concepts and techniques of linear algebra, including: vectors, vector spaces, system of linear equations, matrices, determinants, linear transformations, eigen-theory, diagonalization, Jordan canonical form, matrix exponential, orthogonality, the Spectral Theorem.
Description (Chinese)	本课程旨在介绍线性代数的基础概念和方法。内容包括：向量，向量空间，线性方程组，矩阵，行列式，线性变换，特征理论，对角化，Jordan标准型，矩阵指数，正交性，谱定理。

2. Prerequisites / Co-requisites

A. Prerequisites

None

B. Co-requisites

None

3. Learning Outcomes

- Understanding of linear transformations.
- Proficiency in the basic matrix calculations, such as addition, multiplication, inverse, and determinants.
- Proficiency in matrix operations and solving linear systems.



- Knowledge of matrix decompositions, such as SVD.
- Understanding of basic theory of linear algebra, such as linear independence, vector spaces, bases and dimension, and change of coordinate matrix.
- Proficiency in diagonalization, Jordan canonical forms, and matrix exponential.
- Understanding of important concepts, including orthogonality, self-adjointness, and the spectral theorem.

4. Course syllabus

Vectors: vector operations, linear combinations, linear span, linear independence, spanning sets.

Vector Spaces: vector spaces, subspaces, basis and dimension, change of basis.

Linear Transformations: kernel, image, range, matrix representations of linear transformations, similarity.

System of Linear Equations and Matrices: equivalent systems, equation operations, solution set, augmented matrices, row operations, matrices, vectors, row equivalence, (reduced) row-echelon form, Gaussian elimination, homogeneous system.

Matrices and Linear transformations: column and row spaces, null spaces, rank of matrices.

Determinants: determinants and row operations, minors and cofactors, Laplace expansion, adjoint matrix and matrix inverse, Cramer's rule.

Eigen-Theory: eigenvalues, eigenvectors, trace, characteristic polynomial, diagonalization, simultaneous diagonalization, Jordan canonical form, matrix exponential.

Orthogonality: inner product spaces, orthogonal subspaces, orthonormal basis, the Gram-Schmidt process, self-adjoint operators, the Spectral Theorem, Singular Value Decomposition Theorem.

5. Assessment Scheme

Component/ method	% weight
Assignments (including mini software projects)	25
Mid-Term Exam	25
Final Exam	50

6. Grade descriptor

General



Grade	Description
A	Demonstrates the ability to synthesize and apply the principles or subject matter learnt in the course, to novel situations and/or in novel ways, in a manner that would surpass the normal expectation at this level, and typical of standards that may be common at higher levels of study or research. Has the ability to express the synthesis of ideas or application in a clear and cogent manner.
B	Demonstrates the ability to state and partially apply the principles or subject matter learnt in the course to most (but not all) familiar and standard situations in a manner that is usually logically persuasive. Has the ability to express the knowledge or application in a satisfactory and unambiguous way.
C	Demonstrates the ability to state and apply the principles or subject matter learnt in the course to most (but not all) familiar and standard situations in a manner that is not incorrect but is somewhat fragmented. Has the ability to express the separate pieces of knowledge in an unambiguous way.
D	Demonstrates the ability to state and sometimes apply the principles or subject matter learnt in the course to some simple and familiar situations in a manner that is broadly correct in its essentials Has the ability to state the knowledge or application in simple terms.
F	Unsatisfactory performance on a number of learning outcomes, OR failure to meet specified assessment requirements.

7. Feedback for evaluation

- Collect students' feedback from CTE by the end of the course.
- Collect opinions from peers by internal and external course reviews.

8. Reading

Required Reading

1.Linear Algebra, /, • S. H. Friedberg, A. J. Insel, L. E. Spence, Pearson, 4TH EDITION
PEARSON, United States

9. Course components



Activity	Hours/week
Lecture	3
Tutorial	1

10. Indicative teaching plan

Week	Content/ topic/ activity
1	Linear combination, linear span, linear independence, vector spaces, subspaces.
2	Basis, dimension, coordinates, change of basis, column and row spaces, null spaces, change of basis
3	Linear transformations: properties, kernel, image, matrix representations, change of coordinate matrix, matrix rank.
4	System of linear equations, elementary row operations, (reduced) row echelon form, augmented matrices.
5	Determinants, minors and cofactors, Laplace expansion, properties, adjoint matrix and matrix inverse, Cramer's rule.
6	Eigenvalues, eigenvectors, characteristic polynomials.
7	Diagonalization.
8	Jordan canonical form
9	Matrix exponential
10	Inner product, orthogonal vectors, orthogonal subspaces, orthogonal complement.
11	Gram-Schmidt process, orthonormal basis.
12	Self-adjoint operators, normal operators, simultaneous diagonalization
13	Orthogonal projection, spectral theorem,
14	Singular value decomposition

11. Any other information

12. Version date

Version number	2
As of (date)	2023/12/11